

Section J
Quiz #05

Discrete Structures

Q# $\{-3, -2, -1, 0, 1, 2, 3\}$

$$mRn \iff 3|(m^2 - n^2)$$

(a) All ordered pairs that are part of the relation.

(b) Make a directed graph.

(c) Show

↳ Symmetric

↳ Reflexive

↳ Transitive

(d) $[-3], [-2], [-3]$

SOLUTION

→ all values squared lead to

$$\begin{array}{cccc} (\pm 3)^2 & (\pm 2)^2 & (\pm 1)^2 & 0^2 \\ 9 & 4 & 1 & 0 \end{array}$$

MOD 3

$$\begin{array}{cccc} 0 & 1 & 1 & 0 \end{array}$$

SO

MOD 3 = 0 vals

$$\{-3, 0, 3\}$$

MOD 3 = 1 vals

$$\{-2, -1, 1, 2\}$$

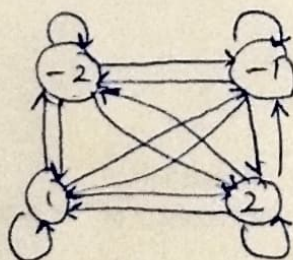
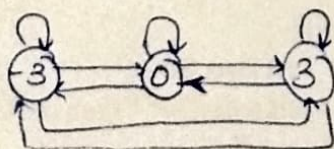
(a)

~~GROUP~~

the group with itself pairs

$$\begin{array}{l} (-3, -3), (-3, 0), (-3, 3), \\ (0, -3), (0, 3), (0, 0), \\ (3, -3), (3, 0), (3, 3), \\ (-2, -2), (-2, -1), (-2, 1), (-2, 2), \\ (-1, -2), (-1, -1), (-1, 1), (-1, 2), \\ (1, -2), (1, -1), (1, 1), (1, 2), \\ (2, -2), (2, -1), (2, 1), (2, 2) \end{array}$$

(b)



(c) Reflexive → True

$$m^2 - m^2 = 0 \text{ and}$$

3|0 is valid.

Symmetric

$$\hookrightarrow m^2 - n^2 \text{ of } m^2 - n^2 - (m^2 - n^2)$$

-K

As K was valid, K is also valid

→ so symmetric

P.S

these proofs aren't done with proper wordings but are correct

Transitive

$$m^2 - n^2 = 3k$$

$$n^2 - p^2 = 3j$$

$$(m^2 - n^2) + (n^2 - p^2) = 3k + 3j$$

$$m^2 - p^2 = 3k + 3j$$

$$m^2 - p^2 = 3(k + j)$$

$$3 | (m^2 - p^2)$$

∴ Transitive

$$(d) [-3] = \{-3, 0, -3\}$$

$$[-2] = \{-2, 1, 0, 1, 2\}$$

$$[-1] = \{-2, -1, 1, 2\}$$